

1 The construct

2 Two uses of L

3 Three readouts agree

one likelihood $P(\text{seq} | \text{structure})$

receptor

binder

complex

Confidence $\phi(P)$

any scalar of the bound-state distribution — blind to binding by construction ($L \perp \phi$, proven)

Leverage L

= mixed 2nd derivative (ablate the partner) — the model's CFG guidance direction

$\propto -\Delta\Delta G_{\text{bind}}$ (binding free energy)

Detect

rank experimental $\Delta\Delta G$ (SKEMPI n=2,948; AB-Bind); zero-shot, competitive with and beyond FoldX physics

Steer

$+\alpha L$ on a frozen, off-the-shelf ProteinMPNN → new interface sequences

controls: random · naive

Inverse-folding judges

ESM-IF1 · MIF · PiFold — leverage ↑

Structure predictors

AF2-multimer · Boltz-2 — ipTM ↑

Physics energy function

FoldX $\Delta\Delta G_{\text{bind}}$ — more favorable ↑

steered L beats random and naive (folds but does not bind) on all three

Confidence is not competence — and it recurs at every level.

an inverse-folding model's confidence and a structure predictor's ipTM are both blind to binding;
the competence always lives in the response to ablating the partner.